Engineering report

Run	run-2026-10-17-01, 2026-10-17T08:00:00Z to 2026-10-17T11:42:00Z
Source	SQL Server: client build 15.0.4385.2 (2019 CU28), sandbox build 15.0.4430.1 (2019 CU32), compatibility level 150
Target	PySpark on Databricks, runtime 17.3 LTS, library migrt_spark_v1_4_3
Versions	lesson_set_client c-3, lesson_set_global g-12, test_definition td-7f3a, tool 1.0.0, validator 1.0.0
Retention	zero retention at Amazon Bedrock: effective retention mode 'none' read through the API. Content never leaves AWS and is never shared with Anthropic.
Model use	287.40 of 400.00 USD; steps by tier {'low': 41, 'medium': 0, 'high': 118}; skipped for budget 0; refusals 1
Single-writer assumption	During a run, converted code is the only writer to the tables it writes. Emulated transactions, identity values and constraint checks are correct only while this holds.

Artefacts

sales.uspCreateOrder (stored procedure)

Status	Verdict (testing module)	Validation (final validator)
Converted	Reconciled	Passed

relies on the single-writer assumption

raised form The order-line loop was rewritten as one set-based step; the faithful form passed the same checks.

Coverage of the original 94%, validator coverage 91%; mutants killed 212, survived 3, not covered 2, equivalent 4; tracing available; ran on sandbox.

Construct	Status	How, or why not
TRY/CATCH with THROW	converted	try/except through the support library
Transaction with ROLLBACK in CATCH	converted with caveat	single_writer
SCOPE_IDENTITY()	converted with caveat	single_writer
Cursor over order lines	raised	LLM step (high tier), 3 candidates, 1 kept

sales.uspArchiveOrders (stored procedure)

Status	Verdict (testing module)	Validation (final validator)
Failed: completeness	Failed	Not run

Failure (completeness): 3 rows that should have been archived were left behind. Layer 4, end-to-end reconciliation; first divergence at statement 7 (DELETE ... OUTPUT INTO); minimised input: 2 orders, cut-off 2024-02-29; seed v-3.

Repair: 3 attempts, exhausted; left for a reviewer.

Coverage of the original 88%; tracing available; ran on sandbox.

Construct	Status	How, or why not
DELETE with OUTPUT INTO	converted	write wrapper
Date boundary on a leap day	failed	completeness

sales.vMonthlyRevenue (view)

Status	Verdict (testing module)	Validation (final validator)
Converted	Reconciled	Passed

pre-existing issue In the original, monthly totals do not add up to the yearly total when an order is refunded in a later month. The conversion reproduces this.

Coverage of the original 100%, validator coverage 100%; mutants killed 64, survived 0, not covered 0, equivalent 1; tracing available; ran on sandbox.

Construct	Status	How, or why not
GROUP BY on a case-insensitive column	converted	an original spelling from each group
DATEADD/DATEDIFF month idiom	converted	date_trunc

ref.uspApplyPriceImport (stored procedure)

Status	Verdict (testing module)	Validation (final validator)
Failed final validation	Reconciled	Failed

fallback A raised form failed final validation; the faithful form was validated again and also failed.

Defect notice: The validator found a difference the testing module missed. The testing module is flagged for a fix; this procedure stays unconverted until a release fixes it.

Coverage of the original 97%, validator coverage 96%; tracing available; ran on sandbox.

Construct	Status	How, or why not
MERGE with WHEN NOT MATCHED BY SOURCE	converted	merge through the write wrapper
Case-sensitive comparison by COLLATE	converted with caveat	collation assumed for one column

sales.uspOrderSummary (stored procedure)

Status	Verdict (testing module)	Validation (final validator)
Blocked by sales.uspArchiveOrders	Blocked by a dependency	Not run

Construct	Status	How, or why not
Call to sales.uspArchiveOrders	blocked	

dbo.uspSearchProducts (stored procedure)

Status	Verdict (testing module)	Validation (final validator)
Contains manual work	Not testable: contains manual work	Not run

manual Full-text search (CONTAINS) is not supported on Databricks.

Construct	Status	How, or why not
CONTAINS full-text predicate	manual work	unsupported on the target
Paging with OFFSET/FETCH	converted	limit and skip

LoadDailySales.dtsx (SSIS package)

Status	Verdict (testing module)	Validation (final validator)
Recognised, not converted	Recognised	Not run

manual SSIS packages are recognised and listed, not converted, in v1.

Degradations (nothing degrades silently)

Kind	Where	What
assumption	ref.Product.Sku	Collation assumed from the database default; the column's own collation could not be read.
trimmed context	sales.uspCreateOrder, raising step	Worked examples cut from 3 to 1 to fit the budget.
unreachable branch	sales.uspArchiveOrders line 41	Proved infeasible by the solver; listed, not counted against coverage.
excluded value	sales.uspCreateOrder	NEWID() output left out of comparison.
volume	run	Tables of 50 rows (validator: up to 5,000); faults that show only at production volume are not tested.

Changes since the last run

- sales.uspCreateOrder: verdict Failed -> Reconciled, because client lesson c-3/L7 (reviewer: J. Doe, delivery engineer)

Open problems for a reviewer

- repair exhausted: sales.uspArchiveOrders
- testing-module defect: ref.uspApplyPriceImport

Required deployment steps

1. Create the counter state table (definition in the project).

2. Run the counter initialisation command at go-live, with the DBA's exported starting values.
3. Run it again after any bulk load that bypasses the write wrappers.